GENOMELENS

Economic analysis of a genomic report

Cost–utility model of acting on the findings in one whole-genome input, valued at $100,000 per QALY from a healthcare sector perspective.

Source input	**public benchmark genome HG003**	Input type	**WGS**
Generated	**24 October 2026**	Time horizon	**each decision model states its own**
Discounting	**3.0%, costs and QALYs**		

Proprietary.
Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.

Public benchmark genome.
A public benchmark genome. It is a real person's genome; the findings are shown to demonstrate the analysis and are not a clinical report.

Research and educational use — not medical or financial advice.
This is a decision-analytic model, not a clinical or economic evaluation submitted to any authority. Its parameters are population averages applied to one genome: it values the expected consequences of acting on a pattern of findings, and does not predict what will happen to any individual. Monetary figures include health valued at a willingness-to-pay threshold and are not cash returned to anyone. Clinically relevant findings require confirmation by an appropriate professional — a physician, clinical pharmacist or board-certified genetic counselor — before anything follows from them.

Economic impact overview

Modeled economic impact

NO HEADLINE VALUE IN THIS GENOME

No primary modeled value to headline: this genome's modeled values are shown below with their class, because their models state no QALY change, which a headline requires; they are measured nulls; they hold only in a stated clinical context. Every other finding is classed with its reason in the table of all economically relevant findings.

1	4	1	13
PRIMARY MODELED	TESTING-STRATEGY / POPULATION-LEVEL VALUE	CONDITIONAL	TRUE ZERO

3	1
UNQUANTIFIED	TECHNICALLY UNRESOLVED

Modeled values (shown separately, never added)

PRIMARY MODELED · ASSUMPTION-DRIVEN SCENARIO

$2,101

lifestyle intervention + medical therapy when a top-decile PRS lifts an intermediate-clinical-risk (FINDRISC ten-to-twenty-percent band) adult to a ten-year type 2 diabetes risk of twenty percent or more (Martikainen strategy). Counted: the selected model's expected value

CONDITIONAL · CONTEXT CONDITIONAL · UNRESOLVED

$1,043

No CYP2C19 loss-of-function allele (normal, rapid or ultrarapid metabolizer) → clopidogrel instead of ticagrelor/prasugrel after primary PCI. Holds only if an acute MI treated by primary PCI in a system whose default is ticagrelor or prasugrel

TRUE ZERO · EVIDENCE-QUALIFIED · TRUE ZERO

$0

CYP2D6 Poor Metabolizer → CYP2D6-guided opioid selection. A measured null: the evidence finds no significant outcome difference

TRUE ZERO · EVIDENCE-QUALIFIED · TRUE ZERO

$0

OPRM1 finding → OPRM1-guided naltrexone. A measured null: the evidence finds no significant outcome difference

SUPPORTED CALCULATION AT THE ASSUMED AGE

Modeled net monetary benefit at assumed age 45: **$0** EVIDENCE-QUALIFIED · TRUE ZERO — the supported calculation at the assumed age: only decisions whose every model input is sourced (here a measured null, which does not depend on the age: the supported decisions show no measured effect on health, and no cost of acting is charged, which is not an unknown). A stricter test, not a total of the values above, and it does not replace them.

Research and educational use only; not medical, clinical or financial advice. Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results.

All economically relevant findings

Every economic result of this genome, ranked within its class. Values in different classes are never added.

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
lifestyle intervention + medical therapy when a top-decile PRS lifts an intermediate-clinical-risk (FINDRISC ten-to-twenty-percent band) adult to a ten-year type 2 diabetes risk of twenty percent or more (Martikainen strategy)		**$2,101**	PRIMARY MODELED **ASSUMPTION-DRIVEN SCENARIO**	Counted: the selected model's expected value
CYP2D6 finding	Pharmacogenomic-guided antidepressant selection (Ghanbarian 2023, 20-year model)	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
CYP2D6 finding	Pharmacogenomic-guided antidepressant selection (PREPARE psychiatry estimate)	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
VKORC1 finding	Genotype-guided warfarin dosing (CYP2C9, VKORC1)	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
Hypothetical, based on assumptions: any adult, before results	Population screening for Lynch syndrome, hereditary breast and ovarian cancer and familial hypercholesterolemia	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
No CYP2C19 loss-of-function allele (normal, rapid or ultrarapid metabolizer)	clopidogrel instead of ticagrelor/prasugrel after primary PCI	**$1,043**	CONDITIONAL **CONTEXT CONDITIONAL · UNRESOLVED**	Holds only if an acute MI treated by primary PCI in a system whose default is ticagrelor or prasugrel
CYP2D6 Poor Metabolizer	CYP2D6-guided opioid selection	**$0**	TRUE ZERO **EVIDENCE-QUALIFIED · TRUE ZERO**	A measured null: the evidence finds no significant outcome difference
OPRM1 finding	OPRM1-guided naltrexone	**$0**	TRUE ZERO **EVIDENCE-QUALIFIED · TRUE ZERO**	A measured null: the evidence finds no significant outcome difference

All economically relevant findings (continued)

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
APOE		**valued at zero**	TRUE ZERO	ACMG/NSGC 2011, reaffirmed: APOE testing for Alzheimer's risk is not clinically recommended
BDNF		**valued at zero**	TRUE ZERO	ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for …
COMT		**valued at zero**	TRUE ZERO	No CPIC guideline doses citalopram or escitalopram by COMT genotype: the CPIC antidepressant guideline's genes are …
CYP2C19 Rapid Metabolizer (RM) (clopidogrel)		**valued at zero**	TRUE ZERO	The genotype does not change care
CYP3A5 Non-expressor (PM) (tacrolimus)		**valued at zero**	TRUE ZERO	The CPIC tacrolimus guideline starts a non-expresser (*3/*3) at the standard recommended dose, so this genotype does …
HLA-DQA1		**valued at zero**	TRUE ZERO	USPSTF 2017: insufficient evidence to screen asymptomatic people for celiac disease (I statement)
HLA-DQB1		**valued at zero**	TRUE ZERO	USPSTF 2017: insufficient evidence to screen asymptomatic people for celiac disease (I statement)
PTPN22		**valued at zero**	TRUE ZERO	No guideline ties an action to a single R620W allele in an asymptomatic adult
UGT1A1 Intermediate Metabolizer (IM) (irinotecan)		**valued at zero**	TRUE ZERO	This phenotype does not change the prescribing decision
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)		**valued at zero**	TRUE ZERO	Not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified …
vitamin D (at age 45)		**valued at zero**	TRUE ZERO	Outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older …

All economically relevant findings (continued)

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
CYP2D6 Poor Metabolizer (PM) (codeine / antidepressants)		**not quantified**	UNQUANTIFIED	Actionable finding, person-level value not quantified: a CYP2D6 poor or ultrarapid metabolizer: CPIC 2023 SSRI ...
Pharmacogenomic-guided antidepressant selection (PREPARE psychiatry estimate) (CYP2D6)		**not quantified**	UNQUANTIFIED	Actionable finding, person-level value not quantified: a CYP2D6 poor or ultrarapid metabolizer: CPIC 2023 SSRI ...
VKORC1 High Warfarin Sensitivity (warfarin)		**not quantified**	UNQUANTIFIED	Actionable finding, person-level value not quantified: the CPIC warfarin guideline (2017, PMID 28198005) gives this ...
NAT2 slow acetylator (three-marker inference): NAT2-guided isoniazid dosing		**not assessable**	TECHNICALLY UNRESOLVED	NAT2 is inferred here from three markers (*5/*6/*7) without phase

The evidence hierarchy

- **Evidence-supported**: every model input is sourced and the value does not rest on an age that is stated rather than observed (badge EVIDENCE-QUALIFIED)
- **Primary modeled**: the selected economic model for a decision; it may depend on the stated assumptions shown beside it **in this genome**
- **Once the decision arises**: the value per person if the decision arises; not an expected value today
- **Alternative modeled**: a competing model or sensitivity scenario for the same decision; never added to the primary result
- **Conditional**: valid only if a stated clinical or treatment condition holds; never added **in this genome**
- **Family / cascade-testing**: accrues to relatives or another beneficiary; never in the individual's personal total
- **Population / testing strategy**: a value per person screened or tested in a population, not this donor's value; never added **in this genome**
- **Hypothetical / projected**: an exploratory scenario on stated assumptions, not an observed or established benefit; in no total **in this genome**
- **Negative**: modeled, and the costs exceed the health gain valued at $100,000 per QALY; shown, never hidden
- **True zero**: the evidence supports no incremental benefit or change in care within the defined decision **in this genome**
- **Unquantified**: potentially relevant, but the evidence is insufficient for a monetary value; not $0 **in this genome**
- **Technically unresolved**: the genomic evidence cannot reliably support the interpretation the decision needs; not $0 **in this genome**
- **Not applicable**: the model or intervention does not apply to the assessed context (sex, age or setting); not $0

Unknown is not $0. Unquantified is not $0. Technically unresolved is not $0. No supported estimate is not a measured zero. Negative is not zero. An alternative is never added.

Findings valued by their decision models

Net monetary benefit (NMB) is the QALYs gained valued at $100,000 each, less the change in healthcare cost, in 2019 US dollars. It is a health-economic value, not cash saved: a positive NMB can come with higher spending. Each clinical decision is valued once; the categories are never added to each other. Each figure's horizon is its own model's (the carrier models run over a lifetime; no pooled disease model ran for this genome).

Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Age 45 is stated for this run, not observed (one demonstration age for every sample; the PGP public profiles give only an age bracket, and the GIAB and 1000 Genomes samples publish no age). A value whose model depends on age is modeled at that age; a value its source gives for one age only is held at that age, and its row says so. Model evidence is stated apart from whether its context matches the donor, and the qualification stays assumption-driven; an age-dependent value would be evidence-qualified only at an observed or self-reported age.

Sex: known male.

Supported calculation at assumed age 45

$0 EVIDENCE-QUALIFIED · TRUE ZERO · AGE 45 ASSUMED every decision supported at the assumed age is a measured null, so the supported calculation is a measured zero. Measured nulls eligible under the available donor context, each with its age restriction, are listed in A. Whether it applies to this donor is not established: the donor's age is not observed.

- no decision is supported at the assumed age beyond its measured nulls

A. Donor-specific economic value

Not established with available context — the donor's age is not observed, and 1 priced decision(s) depend on it; no total is stated for this donor, which is not $0. Results eligible under available donor context, each with its age restriction: genuine calculated values, not the donor's total.

- **CYP2D6-guided opioid analgesia (codeine, tramadol)**: $0 EVIDENCE-QUALIFIED · TRUE ZERO
  measured null in adults with acute or chronic pain: no analgesia benefit (acute pain P = .80; chronic pain p = 0.74); codeine incident use 9.5% over 4 years, insured adults 40-64. The switch is between generic analgesics, so no cost of acting is charged, so no health gain and no cost of acting is credited.
  Age restriction: the chance it arises was measured in insured adults 40-64.
  comparator: usual analgesia; sources: PMID 41719044, PMID 39921243, PMID 27764192.
- **OPRM1-guided naltrexone**: $0 EVIDENCE-QUALIFIED · TRUE ZERO
  measured null in alcohol use disorder: genotype x treatment P = .32, so no health gain and no cost of acting is credited.
  Age restriction: no age restriction stated by the source.
  comparator: naltrexone regardless of genotype; sources: PMID 25760804.

Findings valued by their decision models (continued)

B. Other modeled expected value, with further limitations

Further-limited subtotal $2,101 ASSUMPTION-DRIVEN SCENARIO — limited by inputs besides the age; never added to the supported calculation above.

- **lifestyle intervention + medical therapy when a top-decile PRS lifts an intermediate-clinical-risk (FINDRISC ten-to-twenty-percent band) adult to a ten-year type 2 diabetes risk of twenty percent or more (Martikainen strategy)**: $2,101
  ASSUMPTION-DRIVEN SCENARIO
  ΔQALY not reported separately by the source; ΔCost inside the source's estimate, not reported separately; ΔLY not reported by the source.
  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.
  Holds at the stated age of 45, assumed for this run, not observed.
  Limited by: the score changes this decision; the score changes this decision; age stated, not observed.
  sources: PMID 36186460.

C. Values that hold only in a stated clinical context

- **No CYP2C19 loss-of-function allele (normal, rapid or ultrarapid metabolizer): clopidogrel instead of ticagrelor/prasugrel after primary PCI**: $1,043 CONTEXT CONDITIONAL · UNRESOLVED
  ΔQALY +0.00898; ΔCost -$145 (2019 USD; positive is more spending); ΔLY not reported by the source.
  Holds only if an acute MI treated by primary PCI in a system whose default is ticagrelor or prasugrel.
  Limited by: its benefit pathway is not specified, so it enters no total; source discount rate; QALY per patient tested, not per non-carrier.
  sources: PMID 31479209, PMID 34490590, CMS NADAC 2026-09-23.

D. Alternatives and scenarios, shown separately

- none for this genome

E. Actionable pathways not yet quantified

- CYP2D6 Poor Metabolizer (PM) (codeine / antidepressants) — actionable economically unquantified: a CYP2D6 poor or ultrarapid metabolizer: CPIC 2023 SSRI guideline (Bousman 2023, PMID 37032427; vendored PA166127638, PA166127639) changes the starting dose or drug for paroxetine and vortioxetine, and the CPIC tricyclic guideline for tricyclics (none of them modeled); citalopram is not a CYP2D6 drug; the only economic evaluations (Ghanbarian 2023; PREPARE) are averages per patient tested with a CYP2D6 + CYP2C19 panel, with no result by genotype, so no value is estimated for this genotype (shown only as testing-strategy values).
- Pharmacogenomic-guided antidepressant selection (PREPARE psychiatry estimate) (CYP2D6) — actionable economically unquantified: a CYP2D6 poor or ultrarapid metabolizer: CPIC 2023 SSRI guideline (Bousman 2023, PMID 37032427; vendored PA166127638, PA166127639) changes the starting dose or drug for paroxetine and vortioxetine, and the CPIC tricyclic guideline for tricyclics (none of them modeled); citalopram is not a CYP2D6 drug; the only economic evaluations (Ghanbarian 2023; PREPARE) are averages per patient tested with a CYP2D6 + CYP2C19 panel, with no result by genotype, so no value is estimated for this genotype (shown only as testing-strategy values).
- VKORC1 High Warfarin Sensitivity (warfarin) — actionable economically unquantified: the CPIC warfarin guideline (2017, PMID 28198005) gives this CYP2C9/VKORC1 genotype a different starting dose from the standard dose, but the economic evaluation (Meckley 2010) is an average per patient starting warfarin, every genotype's dose changed, with no result by genotype, so no value is estimated for this genotype (shown only as a testing-strategy value).

Findings valued by their decision models (continued)

E. Actionable pathways not yet quantified (continued)

- NAT2 slow acetylator (three-marker inference): NAT2-guided isoniazid dosing — technically unresolved: NAT2 is inferred here from three markers (*5/*6/*7) without phase, which is not a prescribing-grade genotype, so no drug decision is valued. Missing input: a prescribing-grade NAT2 genotype.

F. Valued at zero: care does not change (a result, not a gap)

- **APOE** — valued at zero: ACMG/NSGC 2011, reaffirmed: APOE testing for Alzheimer's risk is not clinically recommended; the FINGER trial found no APOE effect modification, and communicating genetic risk does not change behavior. Anti-amyloid labels require their own APOE test (PMID 21577118; PMID 29356827; PMID 26979548)
- **BDNF** — valued at zero: ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for (https://ispg.net/genetic-testing-statement/)
- **COMT** — valued at zero: no CPIC guideline doses citalopram or escitalopram by COMT genotype: the CPIC antidepressant guideline's genes are CYP2D6, CYP2C19, CYP2B6, SLC6A4 and HTR2A, and COMT is not among them; a multi-gene panel's benefit belongs to the panel's guideline genes (PMID 37032427)
- **CYP2C19 Rapid Metabolizer (RM) (clopidogrel)** — valued at zero: the genotype does not change care. CPIC CYP2C19-clopidogrel 2022 update (Lee CR et al., Clin Pharmacol Ther 2022, PMID 35034351): standard dose for RM/UM
- **CYP3A5 Non-expressor (PM) (tacrolimus)** — valued at zero: the CPIC tacrolimus guideline starts a non-expresser (*3/*3) at the standard recommended dose, so this genotype does not change the dose
- **HLA-DQA1** — valued at zero: USPSTF 2017: insufficient evidence to screen asymptomatic people for celiac disease (I statement); ACG 2023 recommends against mass screening, HLA typing being used to rule celiac out (PMID 28350936; PMID 36602836)
- **HLA-DQB1** — valued at zero: USPSTF 2017: insufficient evidence to screen asymptomatic people for celiac disease (I statement); ACG 2023 recommends against mass screening, HLA typing being used to rule celiac out (PMID 28350936; PMID 36602836)
- **PTPN22** — valued at zero: no guideline ties an action to a single R620W allele in an asymptomatic adult; USPSTF rates thyroid screening insufficient evidence (PMID 25798805)
- **UGT1A1 Intermediate Metabolizer (IM) (irinotecan)** — valued at zero: this phenotype does not change the prescribing decision
- **ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)** — valued at zero: not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified Benign in ClinVar (Variation 35707) and is a common polymorphism, not a Wilson disease allele (ClinVar Variation 35707).
- **vitamin D (at age 45)** — valued at zero: outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older (LeBoff 2022, PMID 35939577); Endocrine Society 2024 (Demay, PMID 38828931): for adults younger than 50 it suggests against empiric vitamin D beyond the dietary reference intake and against routine 25(OH)D testing

Findings-first economics

No pooled result: no finding enters the pooled model, so there is no combined estimate to state; each finding is valued by its own decision model, by category, on the next page.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

HEALTH GAIN

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

HEALTHCARE COST CHANGE

Not available

no pooled result

MODELED NET BENEFIT

Not available

no pooled result

DECISION UNCERTAINTY

Not available

no pooled result

0

findings carry a standardized economic estimate

35 of the 35 findings shown carry no standardized estimate, and none is assigned one the model cannot support. Separately, 3 clinical decision(s) are valued by their own decision models (2 evidence-qualified, 1 modeled with stated assumptions), by category in the section 'Findings valued by their decision models'; a finding with such a value has its figure there, not here.

APOE ε4 carrier · OPRM1 OPRM1 A118G (mu-opioid… · COMT COMT Val158Met · BDNF BDNF Val66Met · Vitamin D Synthesis Efficiency · PTPN22 R620W carrier · Celiac Disease Susceptibility · Wilson Disease · Celiac-permissive HLA haplotype… · VKORC1 High Warfarin Sensitivity · UGT1A1 Intermediate Metabolizer · Elevated atrial-fibrillation… · NAT2 — NAT2 slow acetylator… · MTHFR 677TT · Elevated type-2-diabetes… · CYP2D6 Poor Metabolizer · CYP3A5 Non-expressor · CYP2C19 Rapid Metabolizer · OPRM1 — Opioid-response… · CRHR1 — Stress-axis variant · Reproductive genetics · Slow-Wave Sleep Depth · and 13 more

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

This run produced no medication-genotype finding with a priced economic pathway; a finding whose decision the evidence values at zero is listed with the true zeros.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk and prevention findings, in detail

No risk or prevention finding of this genome has a standardized estimate, so there is no card to show. The findings are reported without one below, each with its reason on the glance pages; a lower evidence grade changes how a finding should be read, not whether it is shown.

Reported without a standardized value (33): APOE ε4 carrier (elevated…, COMT COMT Val158Met — COMT Val158Met…, BDNF BDNF Val66Met — BDNF Val66Met…, Vitamin D Synthesis Efficiency…, PTPN22 R620W carrier — autoimmune…, Celiac Disease Susceptibility…, and 27 more. Each carries its reason on the glance pages (no change in care, not assessable, unquantified, or reproductive information); no value is invented.

How the findings combine

Taken together, what do these findings mean?

There is no pooled reference case for this genome: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. Findings bearing on one condition are combined here before any value is credited; with nothing in the pool there is nothing to combine, and no figure stands in for the absent result. The findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'.

INCREMENTAL HEALTHCARE COST

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

INCREMENTAL QALYS

Not available

no pooled result

REFERENCE-CASE NMB

Not available

no pooled result

CLASSIFICATION

Not available

no pooled result

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. Standardized scenario at assumed age 45: the donor-specific economic value is not established with available context.

How sure is this, and what rests on judgment?

Uncertainty has two layers: the spread of modeled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across simulations

No probabilistic analysis of the pooled model: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. No probability of cost-effectiveness, interval or one-way swing is stated for it, because there is no pooled strategy to sample. The decision models carry their own parameter uncertainty (2,000 draws from the ranges the sources print, rescaled to a per-carrier value where a source prints another unit); where a decision has a range, its 5th to 95th percentile is printed beside that decision's value.

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Base case: the value the published evidence supports for this genome, or “not estimable” with the reason. Scenario: what GenomeLens computes on its own assumptions, graded by the quality of its inputs, with each assumption numbered in the log below. Scenario figures are not from published data for this genome and are not the evidence headline; the reproductive costs belong to a prospective child and are in no personal total. A row valued by its own decision model repeats that model's value as a pointer; the model's basis (published, or hypothetical) is stated beside the headline. This genome's assessment is incomplete (a finding is actionable but unquantified, technically unresolved, or lacks the evidence to value): that is not a conclusion that the genome has no further economic value.

Table 1. Results by finding

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
This genome, to the person sequenced (each finding once)	**Not established** Incomplete assessment: not zero	**$12,824** Sum of the rows not marked 'not summed': model and scenario values, never this donor's total	**no evidence-supported value; $2,101 from decision models that are not evidence-qualified; $10,723 from 3 projected finding(s); figures marked 'not summed', reproductive values and payer values are excluded**
APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)
OPRM1 OPRM1 A118G (mu-opioid receptor) — OPRM1 A118G — opioid/naltrexone response	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Valued once, by the decision model(s) of this report: 'OPRM1-guided naltrexone' (a measured null) Sources reviewed: PMID 22784013; PMID 33387367; PMID 31961981
COMT COMT Val158Met — COMT Val158Met — psychiatric medication optimization	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change) Sources reviewed: PMID 8807664; PMID 33387367; PMID 36509836
BDNF BDNF Val66Met — BDNF Val66Met — antidepressant response prediction	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)
Vitamin D Synthesis Efficiency — Nutrient-metabolism variant — deficiency prevention	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Care does not change under a guideline (outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older … Sources reviewed: PMID 17634462; PMID 32533132; PMID 35727271
PTPN22 R620W carrier — autoimmune screening awareness	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. Standardized scenario at assumed age 45: the donor-specific economic value is not established with available context.

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 1. Results by finding (continued)

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
Celiac Disease Susceptibility (second-most-common allele) carrier (susceptibility allele)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: not a pathogenic variant
Celiac-permissive HLA haplotype (DQ2.5/DQ8) — serology decision	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)
VKORC1 High Warfarin Sensitivity (warfarin)	**Not estimable** Not estimable	**$1.05** Published population average, applied to this genotype (not summed)	No earlier figure; the decision models' verdict: actionable, with no economic estimate yet: actionable finding, person-level value not quantified: the CPIC warfarin … Source: PMID 20014877
UGT1A1 Intermediate Metabolizer (IM) (irinotecan)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: this phenotype does not change the prescribing decision
Elevated atrial-fibrillation polygenic risk	**Not estimable** Not estimable	**$7,981** Placeholder estimate (assumption)	No base-case figure. The one polygenic score with a per-person, horizon-matched, US-dollar cost-utility analysis is coronary artery disease …
NAT2 — NAT2 slow acetylator — occupational cancer risk awareness	**Not estimable** Not estimable	**Not estimable** Conditional on a genotype call this input cannot establish; no figure is shown	No earlier figure; the decision models' verdict: not assessable from this input: NAT2 is inferred here from three markers (*5/*6/*7) without phase, which is not a … Sources reviewed, none of which carries a figure: PMID:16112301
MTHFR 677TT — methylfolate supplementation	**Not estimable** Not estimable	**$2,721** Placeholder estimate (assumption)	No base-case figure. The governing guideline recommends against ordering this test: the ACMG practice guideline (Hickey et al. 2013, Genet Med, PMID 23288205) finds that … Sources reviewed, none of which carries a figure: PMID 23288205; PMID 12387655
Elevated type-2-diabetes polygenic risk	**See its decision row** Decision model	**$2,101** Decision model (not summed)	Valued by the decision model 'lifestyle intervention + medical therapy when a top-decile PRS lifts an intermediate-clinical-risk (FINDRISC ten-to-twenty-percent band) …
CYP2D6 Poor Metabolizer (PM) (codeine / antidepressants)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: actionable, with no economic estimate yet: actionable finding, person-level value not quantified: a CYP2D6 poor or …

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. Standardized scenario at assumed age 45: the donor-specific economic value is not established with available context.

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 1. Results by finding (continued)

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
CYP3A5 Non-expressor (PM) (tacrolimus)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: the CPIC tacrolimus guideline starts a non-expresser (*3/*3) at the standard recommended dose, so this …
CYP2C19 Rapid Metabolizer (RM) (clopidogrel)	**Conditional** Conditional on a clinical context	**$21** Conditional value times the chance its context arises	'No CYP2C19 loss-of-function allele (normal, rapid or ultrarapid metabolizer): clopidogrel instead of ticagrelor/prasugrel after primary PCI': 1,043.43 USD if an acute … Source: PMID 31479209; PMID 34490590
OPRM1 — Opioid-response pharmacogenomic variant	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Valued once, by the decision model(s) of this report: 'OPRM1-guided naltrexone' (a measured null)
CRHR1 — Stress-axis variant (CRHR1/FKBP5)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care
Reproductive genetics — 3 carrier condition(s) where partner testing would be informative	**Separate arm** Separate arm	**See reproductive table** Reproductive table	Costed per risk in the reproductive rows (a prospective child's value, kept out of the personal total)
Slow-Wave Sleep Depth — Deeper slow-wave sleep (slower adenosine clearance)	**Not estimable** Not estimable	**No model yet** No model yet	No economic model covers this finding, and no figure is assigned in its place.
Stress Resilience (BDNF) — BDNF Met carrier — reduced activity-dependent BDNF	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No economic model covers this finding, and no figure is assigned in its place.
Oxidative Stress Defense — Reduced antioxidant enzyme efficiency	**Not estimable** Not estimable	**No model yet** No model yet	No economic model covers this finding, and no figure is assigned in its place.
ABCA4 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
F11 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
predicted-pathogenic DNAH8 (6:38873014)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic UNC80 (2:209842364)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic PCM1 (8:17937364)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor

Hypothetical, based on assumptions (continued)

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 1. Results by finding (continued)

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
predicted-pathogenic PTPN5 (11:18737918)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic THEMIS2 (1:27879739)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic MICAL2 (11:12220356)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic BHMT2 (5:79083814)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic LINGO2 (9:27950496)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
57 further predicted-damaging variant(s), not listed individually	**Not estimable** Not estimable	**No model yet** No model yet	This genome carries 65 predicted-damaging variants. The economics examines the first 40 and names up to 8 unanchored ones

Table 2. Reproductive risks identified, with their cost (per pregnancy, for a prospective child; ancestry used: European)

REPRODUCTIVE RISK	CHANCE	COST OF THE CHILD'S CONDITION	EXPECTED COST	TEST THAT RESOLVES IT	SOURCE OF THE COST
PTPN22 R620W Broad Autoimmunity Susceptibility (T1D, RA, lupus, Graves'); dominant; carrier	**1 in 2** that a child inherits it, per pregnancy	**$5,000** curated outcome cost for a child with one copy	**Not stated** per pregnancy	**$185** a known-variant test of the pregnancy or the child	GenomeLens registered estimate
HLA-DQB1 DQ8 tag Celiac Disease Susceptibility (second-most-common allele); dominant; carrier	**1 in 2** that a child inherits it, per pregnancy	**$3,000** curated outcome cost for a child with one copy	**Not stated** per pregnancy	**$185** a known-variant test of the pregnancy or the child	GenomeLens registered estimate
ATP7B K832R Wilson Disease (variable); recessive; common polymorphism, not a disease allele	**not stated** of an affected child, per pregnancy	**Not available** this allele is commonly reported as a polymorphism rather than a disease allele (classified as uncertain significance in the carrier catalog): it cannot complete an affected genotype, so neither a chance of an affected child nor a cost of the child's condition is stated; there is no risk for a partner's test to resolve, so none is priced	**—** per pregnancy	**—** No test priced: no risk to resolve	GenomeLens registered estimate

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Methods

How these figures were produced, and how to read them

A decision-analytic model values the expected consequences of acting on each finding in this genome. The figures are model outputs under stated assumptions, kept apart from clinical outcomes.

ANALYTIC FRAME

Perspective	**healthcare sector**
Comparator	**each decision model states its own**
Outcomes	**cost, QALYs, NMB**
Time horizon	**each decision model states its own**
Discounting	**3%, costs and QALYs**
Willingness to pay	**$100,000/QALY**
Currency	**USD, United States**
Uncertainty	**decision models: 2,000-draw parameter uncertainty; pooled model: no PSA (no pooled model)**

HOW TO READ EACH FIGURE

- **Evidence-supported value:** every model input is sourced and the value does not rest on an age that is stated rather than observed. Where the age is stated, as for every public genome, an age-dependent value is a standardized scenario and is labeled so.
- **Decision-model rows:** computed from published inputs with at least one assumption (for a public genome, the stated age); shown, never presented as evidence-qualified.
- **Hypothetical, based on assumptions:** what a finding could be worth if its curated assumptions held; never added to any counted total.
- **Projected, based on assumptions:** the GenomeLens scenario beside a base case that cannot be computed soundly; not the evidence headline.
- **Reproductive costs:** per pregnancy, for a prospective child, never in a personal total.

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labeled as such.
- Costs are restated to 2019 US dollars from each source's price year.
- Findings without a modeled decision pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Proprietary method. Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.